



ARTIFICIAL INTELLIGENCE

Section 2

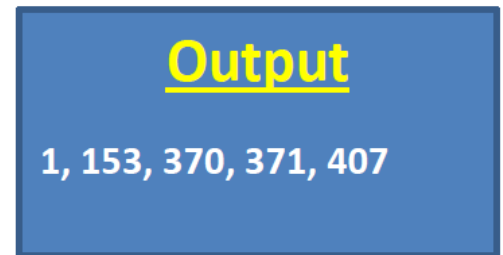
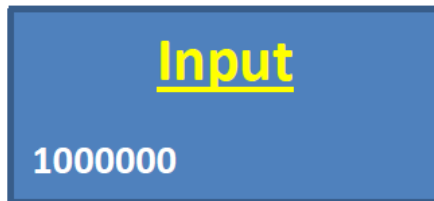
1. Write Python and declare 2 variables (x, y), initialize x = 3 and y = 5 then swap their values.
2. Write Python program that takes an input from user determine the input state:
 - If the input is a number or digit output **"Digit"**
 - If the input is a character print the character **ASCII code** and print his case:
 - If the character is capital output **"Capital"**
 - If the character is small output **"Small"**
3. You have a quadratic equation like this ($ax^2 + bx + c = 0$). You will be given values of (**a, b, c**) write python program that solves the equation using The Quadratic Formula.
If the equation is solvable with real value using the formula print the **equation roots** using the equation:
$$\text{Quadratic Formula} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

otherwise print **"Not solvable"**
4. Write python program that takes an integer and print the **summation of his digits**.
5. Using **For** loop write python program prints the following sequences of values: **20, 14, 8, 2, -4, -10, ..., -994, -1000**
6. Write python program that takes a 2D matrix and finds **the maximum value** and **its location (the row and column indices)**

7. Write python program to print out number of all Armstrong numbers between **1** and **n**. While (n) is a given number.
If sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number.

For example, $153 = (1 * 1 * 1) + (5 * 5 * 5) + (3 * 3 * 3)$

Example 3:



8. Write a python function that takes a list and return **the same list in reverse order**.
9. Write a python function that takes a list and return **the sorted list**.
10. Write python program reads numbers until negative number is entered, display the **number of values entered** and the **largest value which divisible by 3**.
11. Write python program that accepts an integers of size **N**, then prints **each number** and **the number of its occurrences sorted in an increasing order**.

```
Input: 12    5 2 1 6 3 2 1 | 5 2 1 6 4
Output:
1 3
2 3
3 1
4 1
5 2
6 2
```

13. Romania Problem:

Romania Problem Implementation

- **Problem:** On holiday in Romania; currently in **Arad**. Flight leaves tomorrow from Bucharest. Find a short route to drive to **Bucharest**.

- **Start State:**

Arad

- **Goal test:**

Is the current city is Bucharest

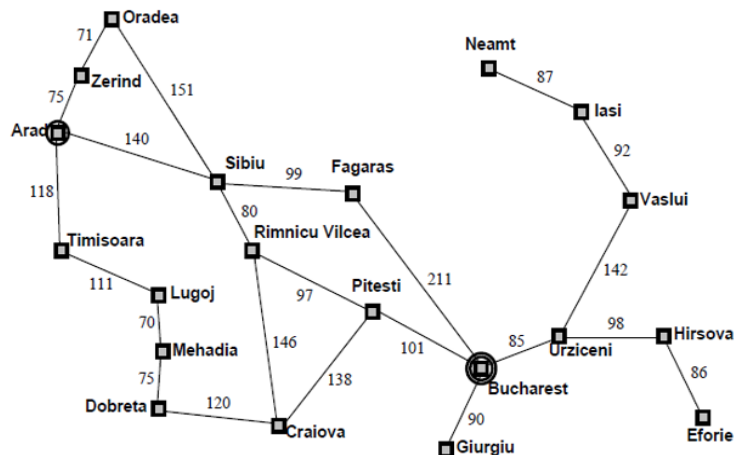
- **Actions:**

Drive between cities

- **Path Cost:**

(Arad, Zerind) = 75

(Arad, Zerind, Oradea) = 146



12. 8-Puzzel Problem:

8-Puzzel Problem Implementation

Start
State

1		2
4	5	3
7	8	6

Goal
State

1	2	3
4	5	6
7	8	

- **Start State:**

Any state can be designated as the initial state. For example, In figure

- **Goal test:**

This checks whether the state matches the goal configuration. For example, In figure

- **Actions:**

{Top, Right, Left, Down}

- **Path Cost:**

1 per action